No torus with ten convex quadrilateral faces

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Abstract

No acoptic polyhedral torus in $\mathbb{R}^3$ has exactly 10 strictly convex quadrilateral faces. This is the $F = 10$ case of Grünbaum–Szilassi Conjecture 3.2; with Seamons's 11-face torus it settles which face counts occur. The proof reduces to the map $\{4,4\}_{(3,1)}$ and certifies, by an LRAT proof checked with the formally verified checker `cake_lpr`, that it has no planar strictly convex realisation, even allowing self-intersections.

1 Introduction

Grünbaum and Szilassi [1] asked which numbers F of strictly convex quadrilateral faces an acoptic polyhedral torus can have (Conjecture 3.2). Seamons realised $F = 11$ (2026), leaving $F = 10$ as the remaining case. This note shows that $F = 10$ is impossible. The certificates and checker are public.

2 Statement

We use the definitions of Grünbaum and Szilassi [1, p. 21].

- Faces are planar simple polygons with no collinear adjacent edges.
- The faces at a vertex form one circuit.
- Adjacent faces are not coplanar.
- The polyhedron is acoptic: every component of the intersection of two faces is a common vertex or a common edge.

Theorem 1. *No acoptic polyhedral torus in $\mathbb{R}^3$ has exactly ten faces that are all strictly convex quadrilaterals.*

Theorem 2. *Let $Q_i = (i, i+1, i+8, i+7)$, indices mod 10, be the faces of the map $\{4,4\}_{(3,1)}$. The map has no realisation in $\mathbb{R}^3$ with planar, strictly convex faces and non-coplanar adjacent faces, even if faces may intersect.*

Corollary 3. *Assume the eleven-face witness of [3] is correct, and assume [1, Prop. 3.1]. Then an acoptic torus with F strictly convex quadrilateral faces exists if and only if $F = 9$ or $F \geq 11$.*

3 Reduction to one map

Lemma 4. *In a torus as in Theorem 1, $V = 10$, $E = 20$ and every vertex has degree 4.*

Proof outline.

- From $4F = 2E$ and $V - E + F = 0$ we get $E = 20$ and $V = 10$.
- Two convex faces have a convex intersection. Being acoptic, it is empty, a common vertex or a common edge.
- At a vertex v of degree d, the points v, its d neighbours and its d opposite corners are pairwise distinct. A coincidence would make some segment vw a diagonal of one face lying in a second face, which the previous point forbids.
- Hence $2d + 1 \leq 10$, so $d \leq 4$. The degree sum is $40 = 4V$, so every vertex has degree 4.

□

Lemma 5. *The face lattice is that of $\{4, 4\}_{(3,1)}$.*

Proof outline.

- A degree-4 quadrangulation of the torus is $\mathbb{Z}^2/L$ with $[\mathbb{Z}^2 : L] = 10$.
- By Lemma 4, $L \cap \{-2, \ldots, 2\}^2 = \{0\}$.
- In Hermite normal form $L = \langle (a, 0), (b, d) \rangle$ with $ad = 10$, this leaves only $L = \langle (10, 0), (3, 1) \rangle$ and its mirror image.
- Two independent exhaustive enumerations confirm this.

□

4 Sign conditions and the SAT certificate

Write $\chi(p_1, \ldots, p_4) = \operatorname{sign} \det[1\ p_i]$. Every realisation satisfies the following.

(GP) For all a, b and all distinct $y_1, \ldots, y_4$, the signs of $[aby_1y_2][aby_3y_4]$, $-[aby_1y_3][aby_2y_4]$ and $[aby_1y_4][aby_2y_3]$ are all zero or contain both signs (the three-term Plücker identity).

(PL) $\chi(Q_i) = 0$ for every face.

(AD) If faces F and G share an edge ab, c is another vertex of F, and x, y are the vertices of G off ab, then $\chi(a, b, c, x) = \chi(a, b, c, y) \neq 0$. Reason: G is strictly convex, and its plane meets plane(F) exactly in the line ab.

Lemma 6. *GP, PL and AD for $\{4, 4\}_{(3,1)}$ are jointly unsatisfiable.*

- The CNF has 420 sign variables and 133 310 clauses.
- CaDiCaL proves it unsatisfiable. The LRAT proof (92 KB) is accepted by the CakeML-verified checker `cake_lpr` and by a second checker.
- A separately written encoder (no auxiliary variables) gives a CNF that is also unsatisfiable, with a verified LRAT proof.
- Both encoders are tested on genuine nine- and eleven-face tori: every clause holds for their exact chirotopes.

- Already the 8 faces left after deleting Q_0 and Q_3 give an unsatisfiable CNF, with verified LRAT proofs from both encoders. Inside it, a solver-reported group-minimal core uses 16 AD conditions and 37 GP relations; its minimality is not certificate-checked.

Lemmas 4–6 prove Theorem 1. Lemma 6 does not use non-intersection, which gives Theorem 2.

5 Data and verification

Code, certificates and a one-command checker: `https://github.com/LeeMarshall1113/ai-math-results`, archived on Zenodo, doi:10.5281/zenodo.23748157 (all versions).

- The SHA-256 hash of the core CNF is `9afbf70b30fd88a8c95839f163f25f1ee21c89f380faf986a1284b239938f4a2`.
- The check needs no human-readable proof of Lemma 6. Such a proof is not available.
- Residual literature risk: Simutis's 1977 thesis was not obtained.

6 Discussion

The argument uses strict convexity of every face. Whether $\{4,4\}_{(3,1)}$ has an acoptic realisation with some non-convex quadrilateral faces is left open.

AI use

This work was carried out with AI assistance (Claude, Anthropic), directed by the author. The computations were independently re-implemented and checked, and the result was attacked by adversarial AI referees whose reports are in the repository. It has not been peer reviewed. The author is responsible for all content.

References

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